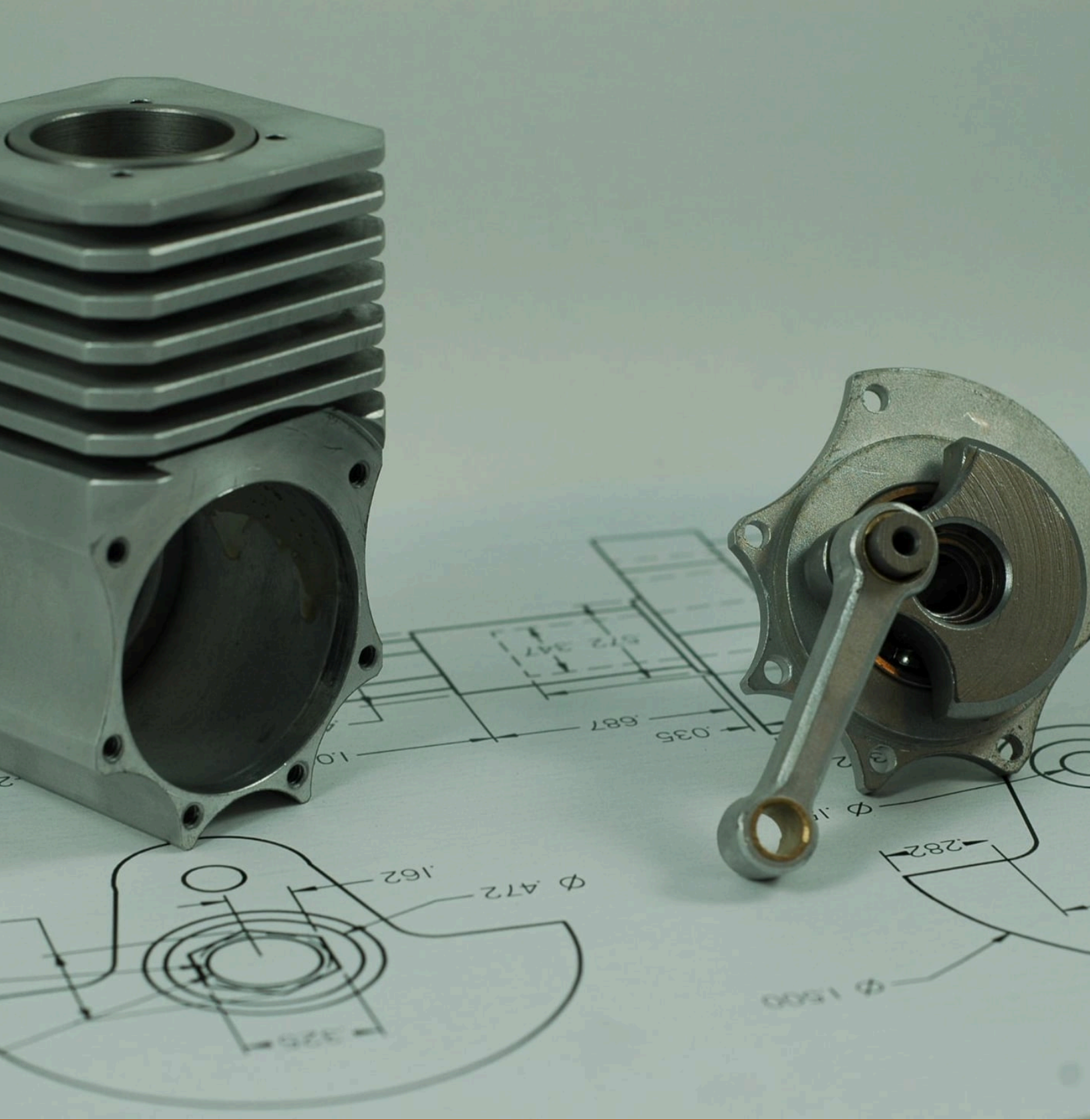




Portfolio

Machinist and mechanic

Contents

Edited Inline	3
2 Cyl RC Crankshaft	6
2 Stroke Cylinders	7
2in Boring Head	8
2017 Africa Twin Valve Clearance Check	9

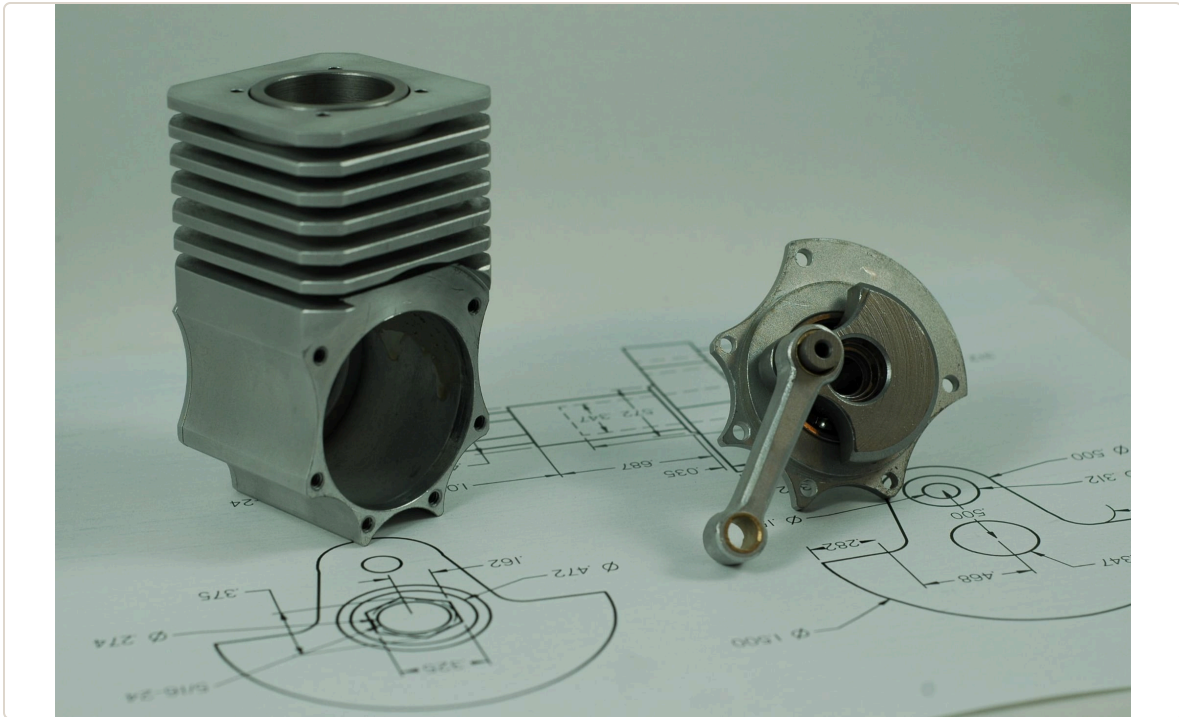
Edited Inline

February 2023 · RC aircraft engine of my own design

1 cubic inch double over head cam model aircraft engine featuring a splash lubricated bottom end, shim-over-valve style valve adjustment, and a timing belt.

ENGINES

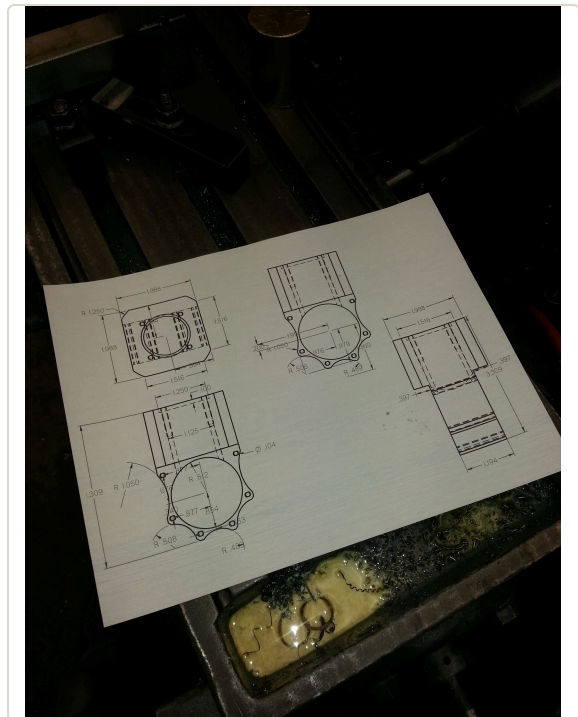
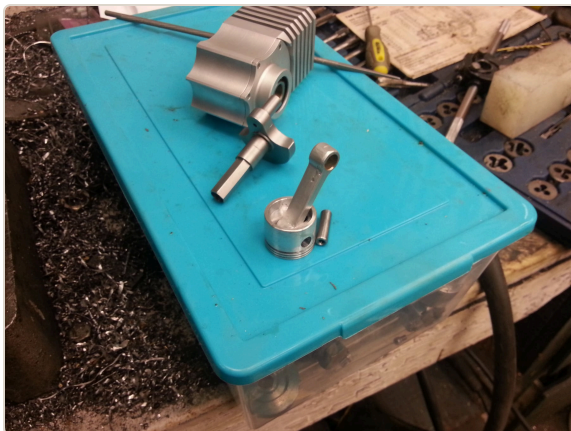
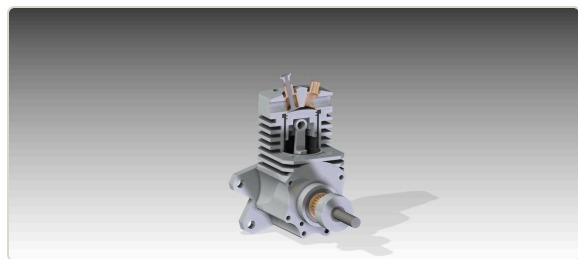
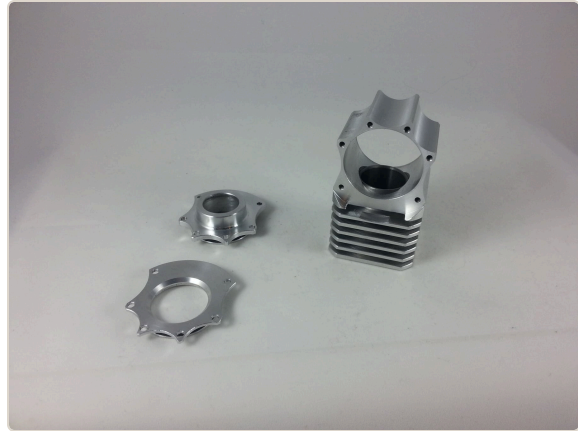
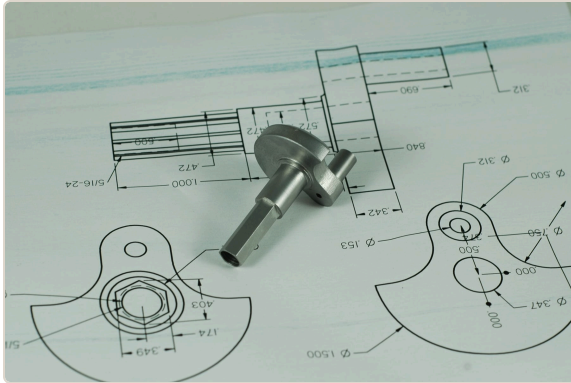
MACHINING

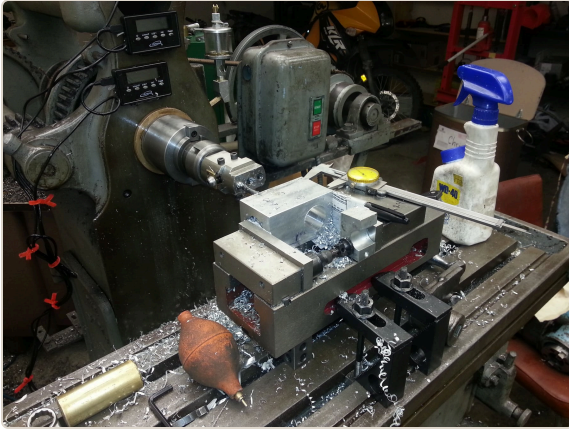


Held flatness under 0.0002 in over 14 in. This is a 1 cubic inch displacement model aircraft engine of my design. It is a 4 stroke, and rare in the model aircraft world, I plan to make it double overhead cam. In a typical 4 stroke RC model engine the valve train uses rockers and pushrods and is driven off of a camshaft that is geared off of the crankshaft. Using Solid Edge, I designed this to be a shim-over-valve double overhead cam, and verified clearances, compression ratio, and valve and ignition timing with Solid Edge motion studies. It will have approximately a 9.1:1 compression ratio, and run on gasoline.

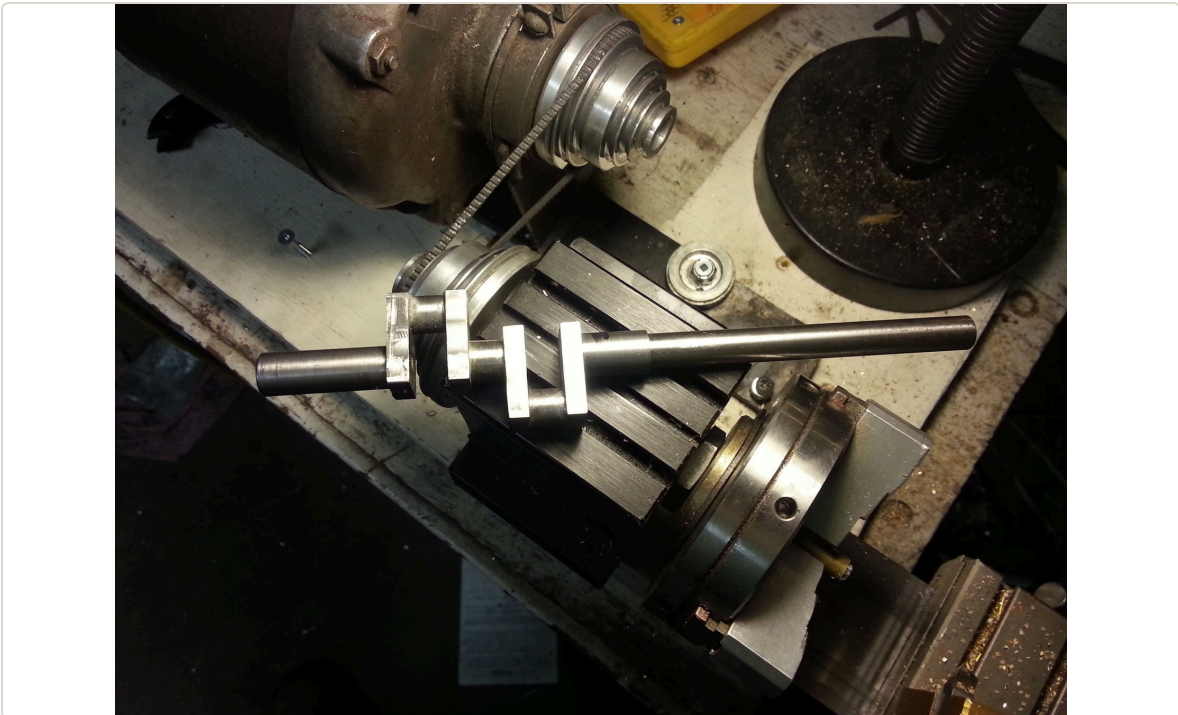
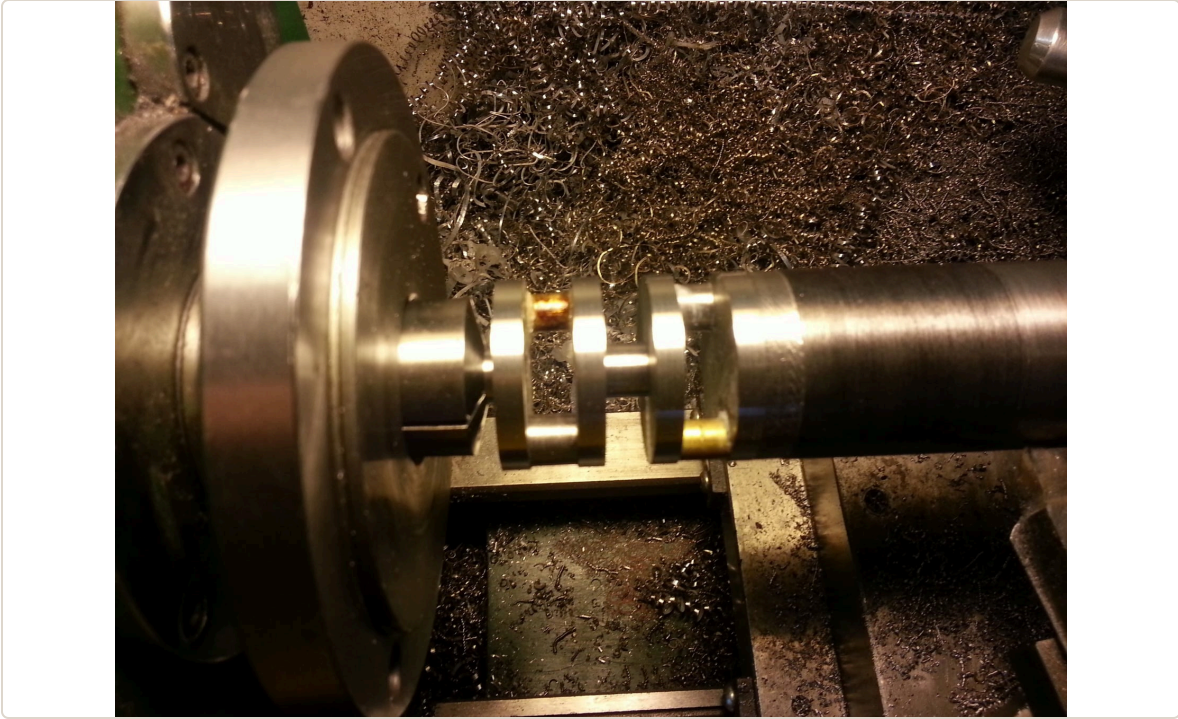
Another unusual thing in the RC engine world is an oil filled crank case. Usually, the bottom end is lubricated using the oil in the fuel by allowing blow by either without piston rings using a lapped fit, or with a single piston ring omitting an oil control ring. This engine has two compression rings and a rudimentary oil scraping ring.

It is about 60% complete. Next to build is the head and the valve train.





2 Cyl RC Crankshaft



2 Stroke Cylinders



2in Boring Head



2017 Africa Twin Valve Clearance Check

Every 24,000 km we get to see some clever engineering

ENGINES

MOTORCYCLES



The 24,000 kilometer service for the first generation Africa Twin involves checking the valve clearances. It sounds like a big job, but it's not too bad. At this point I am practiced enough that I can get access to the valve train in around 1.5 hours, after which it's easy to check.

The Africa Twin has a somewhat unique valve train for being a flagship model adventure bike. Rather than using a double over head cam like most big bikes, it uses a single camshaft over the intake valves only, using a typical shim-over-bucket type adjustment. The exhaust valves are driven off of rockers that ride on that same single camshaft using 4&amp;amp;amp;amp;amp;nbsp; very nice cam follower bearings. The exhaust rockers feature similar adjustment screws that you'd see on a bike from many a yester-year. I think this is a pretty clever design. In most engines, the exhaust valve seats tend to wear and recede faster than the intake valve seats. It seems Honda left easy adjustability for the valves where you would expect there to need adjustment, and saved space on the intake valves where you typically don't ever need to adjust them.

On my specimen after 24,000km the valve clearances were well within spec. At 45,000 when I sold the bike, the exhaust valve clearance had only shrunk by 0.0005". It seems this will be a very long lived engine with little actual adjustment

needed. I wouldn't be surprised if we see later models start adopting 48,000 km valve clearance check intervals like we see in many automotive engines.

There is a trick to the front side fairing pieces: Slide them forwards! If you pry them sideways away from the bike you will break the forward two pins off. Pop the rear of the fairing loose from the tank, and push the fairing forwards to pop the front pins out of the front grommets.

Then the tank and air box need to come off. Take note of the drain tubes running from the left and right rear bottom corners of the air box, you'll need to reattach them when you put this all back together. They can be hard to see and can be easily missed.

